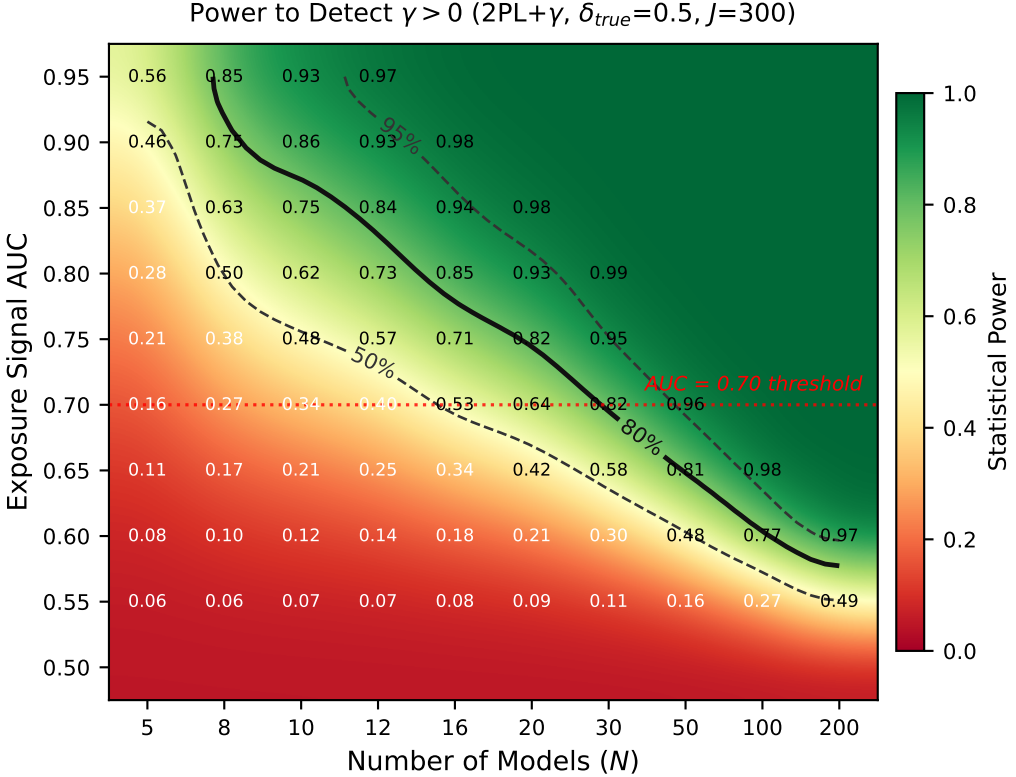

Power to Detect $\gamma > 0$ (2PL+γ, δ_{true}=0.5, J=300)
Exposure Signal AUC
Number of Models (N)
Statistical Power
AUC = 0.70 threshold
95%
80%
50%
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
5
8
10
12
16
20
30
50
100
200
1.0
0.8
0.6
0.4
0.2
0.0
0.56 0.85 0.93 0.97
0.46 0.75 0.86 0.93 0.98
0.37 0.63 0.75 0.84 0.94 0.98
0.28 0.50 0.62 0.73 0.85 0.93 0.99
0.21 0.38 0.48 0.57 0.71 0.82 0.95
0.16 0.27 0.34 0.40 0.53 0.64 0.82 0.96
0.11 0.17 0.21 0.25 0.34 0.42 0.58 0.81 0.98
0.08 0.10 0.12 0.14 0.18 0.21 0.30 0.48 0.77 0.97
0.06 0.06 0.07 0.07 0.08 0.09 0.11 0.16 0.27 0.49